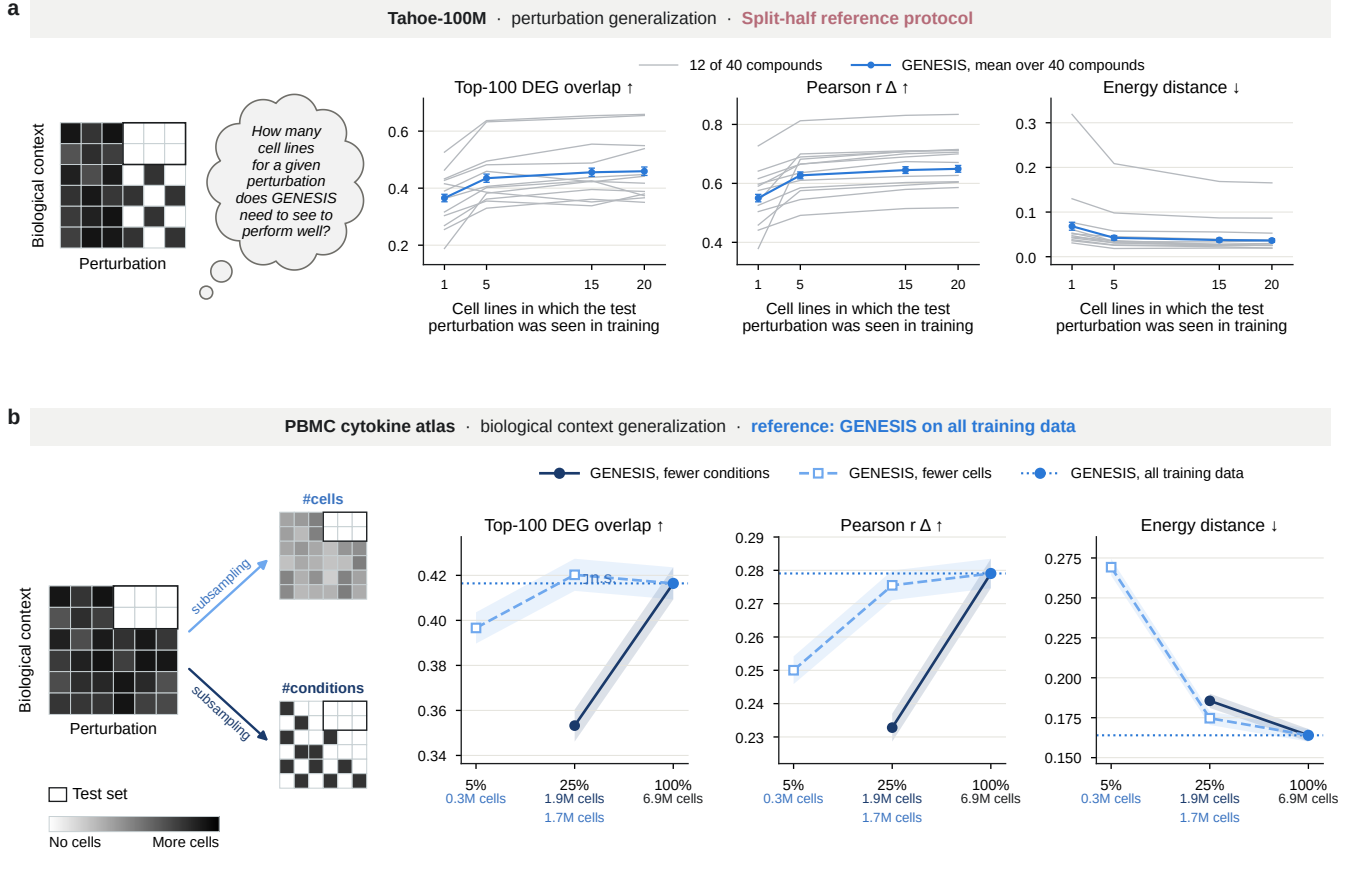
a
Tahoe-100M · perturbation generalization · Split-half reference protocol
12 of 40 compounds
GENESIS, mean over 40 compounds
Biological context
Perturbation
How many cell lines for a given perturbation does GENESIS need to see to perform well?
Top-100 DEG overlap ↑
Pearson r Δ ↑
Energy distance ↓
Cell lines in which the test perturbation was seen in training
b
PBMC cytokine atlas · biological context generalization · reference: GENESIS on all training data
GENESIS, fewer conditions
GENESIS, fewer cells
GENESIS, all training data
#cells
#conditions
subsampling
Test set
No cells
More cells
n.s.
5%
0.3M cells
25%
1.9M cells
1.7M cells
100%
6.9M cells